

## 0.1 Ideal

**Definition 0.1.1.** Let  $R$  be a Ring. A subset  $I \subseteq R$  is called *ideal* of  $R$  if:

1.  $I \subseteq R$  is a subgroup of  $R$ .
2.  $I$  is closed under the multiplication.
3. For any  $r \in R$ ,  $rI \subseteq I$  and  $Ir \subseteq I$ . (In other word, for any  $r \in R, a \in I$ ,  $ra \in I$  and  $ar \in I$ .)

**Theorem 0.1.2.** Let  $R$  be a Ring. Then, TFAE:

1.  $I \subseteq R$  is an Ideal of  $R$ .
2. The additive Quotient Group  $R/I \stackrel{\text{def}}{=} \{r + I \mid r \in R\}$  be a Ring under the operation:

$$(r + I) \times (s + I) = (rs) + I$$

*Proof.* Observation:

$$r_1 + I = r_2 + I \iff r_1 - r_2 \in I \iff \exists a \in I \text{ s.t. } r_1 = r_2 + a$$

Now, for well-definedness, want to show that the equality

$$\begin{aligned} (r + I) \times (s + I) &= (rs) + I \\ &\stackrel{(*)}{=} [(r + \alpha) + I] \times [(s + \beta) + I] = (r + \alpha)(s + \beta) + I = (rs + r\beta + \alpha s + \alpha\beta) + I \end{aligned}$$

(\*) holds for any  $r, s \in R$ ,  $\alpha, \beta \in I$ .

If  $I$  is Ideal, then  $r\beta, \alpha s, \alpha\beta \in I$ . Thus closed under the addition gives (\*).

Conversely, if this operation is well-defined, then for any  $r, s \in R$ ,  $\alpha, \beta \in I$ , (\*) holds.

Substituting zero to each  $r, s, \alpha, \beta$  gives  $I$  is ideal. □

### 0.1.1 Properties of Ideal in Ring with identity

**Definition 0.1.3.** Let  $R$  be a Ring with identity, and  $A \subseteq R$ . Define *Ideal generated by  $A$*  as:

$$(A) \stackrel{\text{def}}{=} \bigcap_{\substack{I \text{ ideal} \\ A \subseteq I}} I$$

And,

$$\begin{aligned} RA &\stackrel{\text{def}}{=} \{r_1a_1 + \cdots + r_na_n \mid n \in \mathbb{N}, r_i \in R, a_i \in A\} \\ AR &\stackrel{\text{def}}{=} \{a_1r_1 + \cdots + a_nr_n \mid n \in \mathbb{N}, r_i \in R, a_i \in A\} \\ RAR &\stackrel{\text{def}}{=} \{r_1a_1r'_1 + \cdots + r_na_nr'_n \mid n \in \mathbb{N}, r_i, r'_i \in R, a_i \in A\} \end{aligned}$$

**Lemma 0.1.4.** Let  $R$  be a Ring with identity, and  $A \subseteq R$ . Then,  $(A) = RAR$ .

*Proof.* Since  $RAR$  is ideal which contains  $A$ ,  $(A) \subseteq RAR$ .

And, conversely, if  $\sum_{i=1}^n r_ia_ir'_i \in RAR$ , then  $\sum_{i=1}^n r_ia_ir'_i \in (A)$  because each  $r_ia_ir'_i$  are contained in  $(A)$ , being  $(A)$  is ideal containing  $A$  and ideal is closed under the addition. □

**Theorem 0.1.5.** Let  $I$  be an ideal of Ring  $R$  with identity.

$I = R$  if and only if  $I$  contains a unit.

*Proof.* Right direction is clear by  $1 \in R = I$ .

Denote  $u \in I$  be a unit with  $vu = 1$ , and Let  $r \in R$  be given. Then,

$$r = r1 = rvu \in I$$

□

**Definition 0.1.6.** An Ideal  $M$  of  $R$  is *Maximal ideal* if: There is no Ideal  $I$  such that  $M \subsetneq I \subsetneq R$ .

**Theorem 0.1.7.** Let  $R$  be a Ring with identity.

Then, every proper ideal  $I \subsetneq R$  is contained in a maximal ideal.

*Proof.* □

Prop. Let  $I, J \subset R$  be ideals of a Ring  $R$  with unity.

1)  $I+J$  is the smallest ideal of  $R$  containing both  $I$  and  $J$ .

2)  $IJ$  is an ideal contained in  $I \cap J$ .

3) If  $R$  is commutative and  $I+J=R$ , then  $IJ=I \cap J$ .

Proof. The first and second, ask a toddler on the street.

Let  $a \in I \cap J$ . Since  $1 \in R = I+J$ ,  $1 = b+c$  for some  $b \in I$ ,  $c \in J$ , so

$$a = a \cdot 1 = a \cdot (b+c) = ab + ac = \underbrace{ba}_{\in IJ} + \underbrace{ac}_{\in IJ} \in IJ.$$

Thus  $I \cap J \subseteq IJ$ .

35. Let  $I, J, K$  be ideals of  $R$ . (with 1)

(a) Prove that  $I(J+K) = IJ + IK$  and  $(I+J)K = IK + JK$ .

(b) Prove that if  $J \subseteq I$  then  $I \cap (J+K) = J + (I \cap K)$ .

**Lemma 0.1.8.** Let  $R$  be a commutative Ring with identity,  $M, P$  are proper ideals of  $R$ .

1.  $M$  is Maximal Ideal if and only if  $R/M$  is a field.
2.  $P$  is Prime Ideal if and only if  $R/M$  is an integral domain.

Summary:  $M$  maximal  $\iff R/M$  field  $\implies R/M$  integral domain  $\iff M$  prime.

*Proof.*

$M$  is maximal  $\iff$  There is no ideal  $I$  such that  $M \subsetneq I \subsetneq R$   
 $\iff$  There are Ideals of  $R/M$  only 0 and  $R/M$   
 $\iff R/M$  is field

$P$  is Prime Ideal  $\iff$  If  $ab \in P$ , then  $a \in P$  or  $b \in P$   
 $\iff$  If  $ab + P = P$ , then  $a + P = P$  or  $b + P = P$   
 $\iff$  If  $\bar{a}\bar{b} = \bar{0}$ , then  $\bar{a} = \bar{0}$  or  $\bar{b} = \bar{0}$   
 $\iff R/P$  is integral domain

□